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Composition, structure and dynamics of Dysart Woods, an old-growth mixed mesophytic forest of southeastern Ohio

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Abstract

Dysart Woods is a 23 ha old-growth remnant of mixed mesophytic vegetation located in southeastern Ohio, USA. A designation of mixed mesophytic for this forest has historically been difficult, in part due to the abundance of white oak (*Quercus alba*); however, the dominance of a variety of other hardwoods prevents a simple oak forest designation. Using two 0.35 ha plots on opposing north- and south-facing slopes, we describe the structure and composition of the overstory, understory, and soils, 30 years after their first examination. In 1970, the woods was dominated by beech (*Fagus grandifolia*), white oak, and sugar maple (*Acer saccharum*) — historically, the three most abundant species in this region. At that time, white oak was only present in the largest size classes, was not regenerating, and was predicted to decline in importance through succession. These patterns continue today suggesting that inferences made via overstory–understory relations in regards to forest succession are relatively robust over this time period. Beech and maple have increased in importance; white oak has decreased in importance due to mortality in the larger size classes and decreasing density due to regeneration failure. Coarse woody debris distributions correlated strongly with living stem species' composition and structure implying an equilibrium balance. CWD volume and frequency were dominated by *Quercus* spp. A detailed analysis of forest health showed that all oak species were in severe decline. The oaks are in a disease decline spiral affiliated with a variety of pre-disposing and inciting factors which include their advanced age (>300 years), their large size (> 100 cm DBH), topography, chronic air pollution, drought, and *Armillaria* root rot fungus. Ca:Al molar ratios in the soil are also extremely low (<1.0) and may be having an additional detrimental effect. All other canopy species appear to be healthy. One of the unusual features of this woods is its relatively diverse and high coverage (up to 90%) understory layer. The herbaceous community was sampled throughout the growing season and found to be markedly dissimilar among sample times and habitat productivity (aspect, soil quality, and light). The role of these factors has not been as well studied for herb communities as it has for tree communities. There appears to be a relatively strong linkage between the overstory regeneration and understory coverage. While a variety of woody seedlings were discovered, most were of shade tolerant species. Only a few small seedlings of white oak were discovered, with none advancing past 30 cm in height, indicating strong competition in the understory. Furthermore, this small remnant forest patch is surrounded by an agricultural and second-growth forest matrix with many non-indigenous plants — none of which have been able to enter the woods, suggesting strong equilibrium stability of these old-growth patches. The hardwood forests of the hills region has been heavily impacted by various human cultures for thousands of years. Dendrochronological analysis of a full basal slab cut from a wind-thrown white oak revealed a fairly active period of fire following European settlement. A lack of fire during the early 1600s to mid 1700s suggests that pre-Anglo fire frequency may have been negligible. There is clearly a continued role for the preservation and study of these old-growth remnants. They

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remain integrally important as we attempt to understand and better manage our remaining anthropogenically disturbed landscape.

Keywords: Forest succession; Stand dynamics; Disturbance; Old-growth; Coarse woody debris; Fire ecology

1. Introduction

Following decisive military displacements of native Americans from the Northwest Territory in 1794 and subsequent treaties, there began a steady stream of Anglo immigrants flowing west in to the Ohio Valley in sufficient numbers that Ohio claimed statehood in 1803 (Walker, 1983). The early European settlers found forests that were 'well timbered' with 'a great deal of extraordinarily fine trees, consisting of white and black oaks, red, Spanish & chestnut oaks, sugar maple, hickory, wild cherry, black and white walnut' along with a mixture of other species that were 'plenty and large' such as 'beech, locusts, ash, chestnut, and the cucumber tree' (Walker, 1983) — many of which exceeded 100 cm in diameter, especially in more mesic areas (Gordon, 1969). Realistic estimates suggest that as much as 95% of what is now the state of Ohio was once covered with forest and punctuated by prairies and large openings created by tornadoes (Trautman, 1977). The influence of native Americans on the natural vegetation of the eastern deciduous forest is conflicting (Day, 1953; Russell, 1983; Patterson and Sassaman, 1988). The status of Amerindians in Ohio is likewise conflicting. Some indicate little to no effect, while others discuss widespread clearing and cultivation around rivers and the use of fire to drive game from the woodlands (Gordon, 1969).

The enormously rich timber resource, along with fertile soils and immense game populations, was largely responsible for the rapid settlement of the Ohio Valley (Trautman, 1977). Europeans generally cleared the land for settlement and agriculture through a combination of girdling and fire. Subsequent anthropogenic disturbances resulted from large scale logging, mining, and charcoal production beginning in the mid-1800s and dramatically changed the landscape of southern and eastern Ohio. Today, few remnants of the original vegetation remain. Most exist as small tracts or fragments that were protected by individual landowners or escaped due to surveying errors (Auten, 1941; McCarthy, 1995).

These small tracts of remnant vegetation serve as the only link to the past extensive forests which once dominated the central Appalachians and the state of Ohio. Recently, in large part driven by the biodiversity crisis and concerns about species extinctions and species assemblages, there has been a renewed interest in these old-growth remnants (Wilcove, 1987). Due to their structure, composition, and overall rarity in the landscape, old-growth remnants may harbor species that are not found elsewhere in the region. From an ecological theory perspective, old-growth is often used as the terminal point for most successional models of forest development (Bormann and Likens, 1979; Oliver and Larson, 1990). These stands have an origin, structure, and dynamic dissimilar to most of the surrounding landscape matrix. Perhaps more importantly, old-growth remnants serve as a benchmark, or experimental control, for the comparison of anthropogenically disturbed and actively managed forests which now dominate the landscape (Whitney, 1987; Juday, 1988; McCarthy, 1995).

There have been numerous studies which have examined the composition and structure of old-growth forests in the central Appalachian region (e.g., Martin, 1975; Muller, 1982; Parker et al., 1985; Abrams and Downs, 1990; McCarthy and Bailey, 1996) and Ohio in particular (Bryant, 1987; Boerner and Cho, 1987; McCarthy et al., 1987; Boerner and Kooser, 1991). Through the careful study of these forests we have gained considerable insight in to the nature of these rare ecosystems. Ultimately, we have made sound progress at meta analysis and have now established general criteria or characteristics for the identification of old-growth in the region (Parker, 1989; Martin, 1992; McCarthy, 1995). However, these old-growth remnants continue to be exceptionally important, and many additional questions remain to be asked. For example, there have been relatively few comparative (Goebel and Hix, 1996) or longitudinal studies of old-growth itself. Oddly, old-growth continues to be retained as a relatively stable terminal position in many models of forest succession even though most forest researchers now regard old-growth to be in a dynamic equilibrium (Huston, 1994). We know very little of the mid- to long-term dynamics of these 'terminal' entities in the absence of anthropogenic or stand renewing disturbances. Likewise, the natural disturbance regime is often not well understood, particularly the frequency of fire (Lorimer, 1993). Additionally, in highly dissected topographies, slope aspect is well known to influence stand composition and structure (Wolfe et al., 1949; Cantlon, 1953; Carmean, 1965; Hutchins et al., 1976; McCarthy et al., 1984, 1987) yet few studies have examined this topographic influence in old-growth (McCarthy et al., 1987) or on other stand properties such as ground layer vegetation (Olivero and Hix, 1998) and coarse woody debris (Muller and Liu, 1991). Lastly, while old-growth is believed to be a potential reservoir for biological diversity and approximate a dynamic steady state condition with respect to ecosystem stability (Odum, 1969; Martin, 1992; McCarthy, 1995), few investigations have quantitatively examined either the ground layer plant diversity (Davison and Forman, 1982; Olivero and Hix, 1998) or invasion potential of these small old-growth remnants with respect to non-indigenous species.

In this study, we use Dysart Woods, an old-growth mixed-mesophytic remnant of unglaciated southeastern Ohio, to examine present composition and structure (both in the overstory and understory), estimate general patterns of stand change and regeneration over a 35 year period, assess the relationship between slope aspect and structure and diversity, and to evaluate the resiliency and stability of the understory flora. Dysart Woods is the only identified old-growth forest in the state of Ohio which could be described as mixed mesophytic in nature and likely represents the regional forests at the time of early settlement as described by Walker (1983).

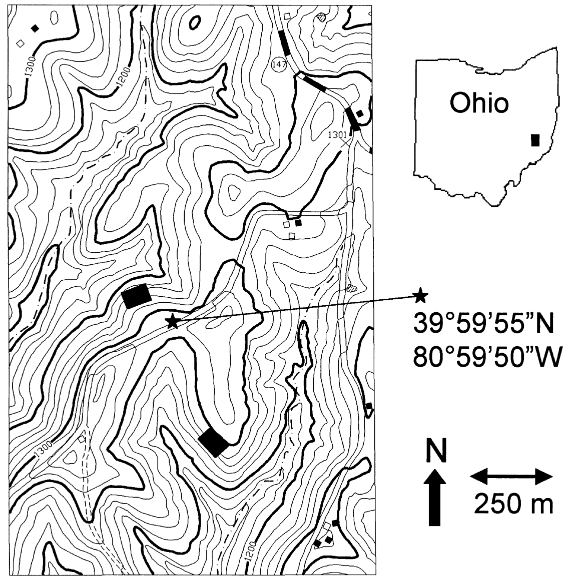

Fig. 1. Portion of United States Geological Survey (1972) 7.5″ Armstrong Mills, OH quadrangle encompassing the old-growth areas of Dysart Woods. An inset map of the State of Ohio shows approximate location of the area. The black rectangles represent the 50 m×70 m permanent plots; upper plot is north–northwest-facing ('north site') and the lower plot is southwest-facing ('south site'). Latitude/longitude co-ordinates are provided from along Ault Dysart road which bisects the area along a ridge. Note S.R.-147 in upper right hand corner of map. All elevations are in feet.

2. Study site

2.1. History

Dysart Woods is a 205 ha land laboratory owned and managed by Ohio University. It is located in eastern Ohio on the unglaciated Allegheny plateau (39°59′55″N, 80°59′50″W, United States Geological Survey, 1972; Fig. 1). Artifactual evidence is available to suggest that Archaic Indians have been present in the immediate area for at least 10,000 years, since the last periglacial environment. The Adena and Hopewell cultures (3000–1000 YBP) were known to be extremely abundant in the region and Indian Mounds are recorded just to the South of Dysart Woods (Knepper, 1989). Unfortunately, the influence on the environment of the Adena and Hopewell peoples, and later Woodland cultures, is not completely understood. Some (e.g., Denevan, 1992) argue that the influence may have been profound and extensive. Gardening occurred in the floodplain river valleys and hunting in the upland areas across the landscape of southeastern Ohio.

Post-Columbian Anglos have a relatively well recorded history in this region and have had a large influence on the land. They were primarily responsible for the conversion of forested land to agriculture and development. The first recorded Anglo farm established in the region was 1788 (Caldwell, 1880). By 1810, 9 years after Belmont was officially declared a county, the region was declared 'settled' and the official US census recorded a population of 88. The first recorded settlers on the land which presently makes up Dysart Woods occurred in 1813 by Miles Hart (Wistendahl, 1971). He and his descendants cleared only the upper slopes and ridges and farmed the land for many years. In 1902 the site was acquired by Henderson Dysart who constructed a farmhouse from local wood, but chose to preserve the magnificent stand of trees which grew on the slopes and broad ravines of the property (Wistendahl, 1971). With the exception of the occasional dead tree near the periphery, the Dysarts did no cutting in these woods; on rare occasions stray cattle may have entered the woods (Lafer and Wistendahl, 1970). A detailed analysis of the ecological disturbance history of these woods has been recently evaluated using dendroecological techniques (Rubino and McCarthy, manuscript).

Due to its uniqueness in the landscape, the property was acquired by the Ohio Chapter of The Nature Conservancy in 1962 from descendants of the Dysart family and later deeded to Ohio University in 1967. The subsequent recognition of Dysart Woods as a National Natural Landmark by the United States Department of the Interior extends the significance of the woods beyond local and state interest. Today, there are 23 ha of old-growth forest contained in three adjacent tracts, with the remaining 182 ha distributed in annually mowed meadows, early successional old-fields, and second-growth forest.

2.2. Vegetation

In 1804, Joseph Gibbons traversed the region throughout Belmont County and proceeded southwards through Washington County (Walker, 1983). He may provide one of the earliest accounts of the vegetation of the region where he describes the forests at length. He recognized the strong topographic influences of the highly dissected topography and spoke of the diversity of species which inhabited the mesic coves and ravines, and the oaks and hickories which dominated the upper slopes. He was particularly impressed by the abundance of sugar maple (*Acer saccharum*) in the region. Likewise, Atwater (1818) described the early forests of Belmont County as being abundant with white oak (*Quercus alba*), sugar maple, and tulip poplar (*Liriodendron tulipifera*). Sears (1925) first mapped the virgin vegetation of Ohio using original land surveys and showed oak (*Quercus* spp.) and beech (*Fagus grandifolia*) as the primary dominants in the area. In mapping the natural vegetation of Ohio, Gordon (1969) mapped the area of Dysart Woods as a White Oak Type under Mixed Oak Forest. However, his description of the White oak–Beech–Maple type as an impoverished subtype of the Mixed Mesophytic Forest previously described by Sampson (1930) more accurately reflects the composition of Dysart Woods. In her seminal work, Braun (1950) included Dysart Woods within the Low Hills Belt section of the Mixed Mesophytic Forest region. She recognized this as a complex vegetation zone with a strong topographic influence with oaks and hickories dominant on the ridges and mixed mesophytic vegetation in the coves and ravines.

In short, the area has proven to be a bit difficult to characterize using standard vegetation protocols due largely to the 14 species which attain dominance or co-dominance in the canopy. Certainly, white oak, beech, sugar maple, and tulip poplar are the major dominants. The largest trees in the forest are white oaks with many stems exceeding 100 cm DBH (diameter at breast height), 37 m in height, and 350 years of age. But, there is also an important admixture of red, black, and scarlet oaks (*Q. rubra*, *Q. velutina*, *Q. coccinea*), black gum (*Nyssa sylvatica*), red maple (*A. rubrum*) and minor contributions by hickory (*Carya ovata*, *C. cordiformis*), black cherry (*Prunus serotina*), white ash (*Fraxinus americana*), and slippery elm (*Ulmus rubra*). A mixed mesophytic designation seems most appropriate for this forest.

2.3. Geology, soils, and climate

Dysart Woods lies in the unglaciated Allegheny plateau physiographic province (Fenneman, 1938), which comprises 25% of the state's southeastern corner and is the only section not ravaged by Pleistocene glaciation (Peacefull, 1996). Geologically, the bedrock is of Pennsylvanian or Permian origin. The topography is highly dissected and considered generally rugged; it contains many hills with long, steep sides and narrow sloping ridgetops. Elevations in the immediate area of Dysart Woods range from ca. 325–425 m.

Soils of Dysart Woods can be classified as Alfisols. These are moderately- to well-drained loam or silt loam surface soils with silty clay, loamy clay, or clay subsoils. They are classified as fine, mixed, mesic Typic Haplaudalfs, primarily of the Lowell–Westmoreland series (Rubel et al., 1981). The Lowell series consists of deep, moderately well-drained, slowly permeable soils on uplands. These soils formed in residuum of limestone, siltstone, and shale with thin layers of sandstone. Likewise, Westmoreland soils are deep but tend to be more permeable and better draining due to their formation from sandstone. The A-horizon ranges from 10–20 cm and the O-horizon may be deep (10–20 cm), especially in the lower ravines and north facing slopes of Dysart Woods. Bedrock is generally around 100 cm (Rubel et al., 1981).

Climate at Dysart Woods is classified as continental (Rubel et al., 1981) with a mean yearly temperature of 10.7°C (National Climatic Data Center/National Oceanic and Atmospheric Administration, 1999). July is the warmest month (mean=22.7°C), and January is the coldest (mean=−1.7°C). Mean annual precipitation is 101.2 cm; July is the wettest month (mean 11.1 cm of precipitation), and October is the driest (mean=6.2 cm). Precipitation is normally abundant and fairly evenly distributed throughout the year with most falling between May and August (National Climatic Data Center/National Oceanic and Atmospheric Administration, 1999). Ohio does not have a distinct dry season (Schmidlin, 1996). Average annual winter snowfall in Belmont County is ca. 80–100 cm, but temperatures rarely remain below 0°C for more than a week, and thus snow does not accumulate. Wind speeds are usually light. While Ohio lies at the northeastern edge of 'Tornado Alley,' winds in excess of 150 km/h are extremely rare in the hilly region of southeastern Ohio (Schmidlin, 1996). Thus, large-scale, stand-renewing wind disturbance is quite unlikely here.

3. Methods

3.1. Permanent plots

The two largest stands of old-growth at Dysart Woods are each ca. 10 ha in size and lie on opposite sides of a ridge, along which runs a township road. The area along the road is disturbed and shows past signs of apple trees and barb wire fencing. This upper slope area was undoubtedly cleared by the Dysart's for agriculture and grazing. As far as can be determined, both stands have an identical past history of ownership and disturbance. One of the goals of this study was to establish long-term permanent plots to study long-term forest stand dynamics and conduct environmental monitoring. Because of a hiking trail which traverses each stand, the largest plot configuration that could be installed was 50 m×70 m (0.35 ha). Each permanent plot was located in an area that remained as free from the influence of the trail as possible. The 'N-plot' was located in the stand to the north of Dysart road on a steep NW-facing linear slope at an elevation of 400 m (Fig. 1). The 'S-plot' was located in the stand to the south of Dysart road on a steep slightly convex SSW-facing slope at an elevation of 410 m (Fig. 1). Steel reinforcing rod was used to mark the plot corners and the internal corners of all 35, 10 m× 10 m subsections.

3.2. Fire history

A basal slab was cut from a recently wind-thrown tree (adjacent to S-plot) and was prepared for tree-ring analysis according to standard dendrochronological procedures (Stokes and Smiley, 1968). Once dates were assigned to individual tree rings and verified (Rubino and McCarthy, manuscript), historical fire events were identified. A fire scar was defined as the presence of charcoal in a growth ring and/or the presence of callus tissue (Abrams, 1985; Smith and Sutherland, 1999). The location of the scars in the rings was noted and fire events were identified as spring or summer/fall if they occurred within the earlywood or latewood, respectively.

3.3. Living vegetation

Within each 0.35 ha plot, all woody stems ≥2.5 cm diameter at breast height (DBH) were identified to species and marked with a numbered aluminium identification tag. Stems were subsequently divided into two structural classes for analysis: saplings (DBH≥2.5 and <10 cm) and trees (DBH≥10 cm).

To characterize the understory (seedlings and herbs), 35 nested circular quadrats (2 and 5 m^2, respectively) were established equidistantly throughout the study plot at regular intervals (i.e., center of 35 sub-plots). The 2 m^2 plots were used to collect cover (%) data for herbs and density (stems ha^{-1}) data for woody seedlings $\leq$30 cm height (Class-I seedlings). The 5 m^2 plots were used to collect density (stems ha^{-1}) data on woody stems<2.5 cm DBH and >30 cm height (Class-II seedlings). All vegetation data were collected from May to August of 1996. Nomenclature follows Gleason and Cronquist (1991).

Vegetation was assessed by standard descriptors of density, basal area or cover, frequency, and importance (Mueller-Dombois and Ellenberg, 1974; Bonham, 1989; Cox, 1990). For the tree and sapling stratum, a relative importance value (RIV; 0–100%) was calculated based on the average of the sum of the relative density (RDEN; %) and relative dominance (RDOM; %). For the woody seedling stratum, DEN (stems ha^{-1}) was used as the measure of abundance. Because herbs are seasonally variable, the herb layer was assessed using cover (COV, %) and relative cover (RCOV, %) as the measure of abundance at three sample periods (mid-April, -June, and -August) to capture spring ephemerals and late season graminoids and composites. In addition, standard estimators of species diversity (richness, Shannon–Weiner, evenness, and Brillouin) were also calculated (see Magurran, 1988 for equations). Percent similarity (community coefficients) was also calculated (using MVSP; Kovach, 1999) among the three sample periods within each aspect to better understand the role of phenological partitioning of these communities.

3.4. Declining, dead, or downed vegetation

Because many of the white oaks in this forest, particularly those in the oldest cohorts which exceed 100 cm DBH and 300 years of age, have obvious decline symptoms appearing (e.g., branch dieback, limb loss, mortality) we also evaluated forest health via symptoms of tree decline. The proximal reasons for this decline are not entirely evident. Certainly these trees may be approaching their maximum life span under forest conditions and the observed mortality may be natural in origin. Alternatively, these trees may be suffering premature dieback and mortality associated with various triggers or stress syndromes associated with climate (drought), pathogens (*Armillaria* spp.), or air and soil pollution (McClenahen and Dochinger, 1985). There are a variety of predisposing, inciting, and contributing factors (Manion and Lachance, 1992) which could lead to increased white oak mortality at Dysart Woods. In an effort to quantify the health of the forest and dominant species, we employed the forest health rating system designed by Millers et al. (1991). Using this system, we rated every stem in the 0.35 ha plots $\geq$2.5 cm DBH for overall crown health and tree vigor. Briefly, this system evaluates: (1) overall appearance and vigor of the tree (healthy to light, moderate, severe decline, or dead), (2) the health and condition of the bole (damage type, causal agent, location), (3) the health and condition of the crown (dieback %, crown transparency, epicormic branching, leaf discoloration or dwarfing, early or late defoliation). The decline index (DI) used by McLaughlin et al. (1992) is an appropriate measure of decline and has been used previously to evaluate the health of a second-growth forest in southeastern Ohio (Walters and McCarthy, 1997). However, we modified this index slightly because we found branch dieback and crown transparency to be the two most useful estimators of hardwood health. We calculated DI as:

$$\mathrm{DI} = \left(\frac{\mathrm{DB}}{2}\right) + \left(\frac{\mathrm{CT}}{2}\right) + (\mathrm{A} * \mathrm{DC}) + (\mathrm{A} * \mathrm{EB})$$

where DI is equal to decline index (%); DB to branch dieback (%); CT to crown transparency (%); DC to discoloration of leaves (%); EB is equal to epicormic branching (%) and A to a weighting factor, $(100-\mathrm{DB})/400$.

The DI is calculated on a tree-by-tree basis and averaged across a species; it ranges from 0 (healthy tree, no decline symptoms) to 100 (dead). A DI<11 is healthy, 11–15.9 has trace decline, 16–20.9 low decline, 21–24.9 moderate decline, >25 severe decline. Only the data for dominant and co-dominant crowns are presented.

To characterize the dead, woody vegetation, we identified, tagged, and measured the length and diameter of all logs $\geq$10 cm basal diameter. Additionally, we measured and tagged all stumps and snags (standing dead trees) $\geq$10 cm basal diameter. To evaluate decay dynamics, all logs, stumps, and snags were given a decay rating from I–V (low to high decay) as described in McCarthy and Bailey (1994) and were identified to the lowest possible taxonomic rank using Panshin and deZeeuw (1980).

To create a permanent map of the study plot, all woody stems, alive or dead, $\geq$2.5 cm DBH were mapped onto an X-Y co-ordinate space using a digitizing tablet and AutoCAD (1996) software. Basal area of trees and snags were illustrated graphically by using four different sized circles to depict trees in roughly 25 cm quartiles. In addition, all downed logs were mapped but relative sizes were not indicated so as to decrease the complexity of the final figure.

3.5. Soils and light environment

To characterize the soil texture and chemistry we collected two bulked soil cores at each of the 35 seedling/herb sample points (at the center of each 10 m$\times$10 m sub-plot). Protocols generally follow those of McCarthy (1997). Soils were returned to the lab at Ohio University and analyzed for percent sand, silt, and clay employing the methods of Bouyoucos (1951). Ash content of the soil (percent organic matter) was evaluated by loss on ignition at 550°C. Fresh soil was used to evaluate pH using a 1:1 soil: water mixture and a Corning 340 digital meter standardized at pH 4.0 and 7.0 (Carter, 1993). Nitrate nitrogen was evaluated using fresh soil and the colorimetric procedures of the Hach Co. (1990). All soils were then dried at 105°C for 72 h and ions extracted using a Mehlich-III extractant (Carter, 1993). A Varian™ 20 atomic absorption spectrophotometer was used to determine the amount of aluminum, calcium, magnesium, and potassium based on absorption or emission as appropriate. Standard colorimetric methods and a Spec-20™ spectrophotometer equipped with far red wavelength capacity were used to assess phosphorous (Carter, 1993).

The understory light environment was characterized using hemispherical photography following the methods of Robison and McCarthy (1999). Briefly, a hemispherical photograph was obtained at the corner of each 10 m$\times$10 m subsection; thus N=48 for each plot. A Sigma 8 mm lens was used on a Nikon camera with a tripod set at 1.5 m. Kodak™ Ektachrome ASA-400 film was used to capture the images, and images were subsequently digitized. All post-processing was accomplished using Canham's (Canham, 1995) GLI/C (v. 3.1) computer software. Percent global light, which averages the effects of direct beam radiation and indirect radiation across the growing season, was calculated.

4. Results

Tree-ring analysis revealed 27 fire events (Fig. 2) from 1624 to 1997. The first fire scar was observed in 1731, and the last scar was noted in 1881. The median fire return interval during this period was 2.0 years. Twenty (74.1%) of the fires were observed during the spring, and seven (25.9%) occurred in the summer. Since all fires preclude available climatic records, we could not correlate fire occurrence with periods of low precipitation and high temperature (sensu Sutherland, 1997). However, reconstructed Palmer Drought Severity Index (PDSI) values (Palmer, 1965; Cook et al., 1999) are available for the fire period. Fire scars resulting from spring fires were observed in years with reconstructed PDSI values ranging from −2.94 (moderate drought) to 1.24 (slightly wet) while

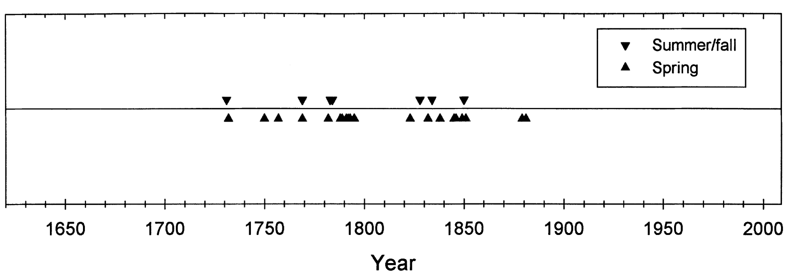

Fig. 2. Historical fire distribution at Dysart Woods. Data are based on dated fire scars observed in a basal slab cut from a 324-year-old, windthrown white oak tree.

summer/fall fires were observed in years with reconstructed PDSI values that ranged from −1.53 (mild drought) to 1.49 (slightly wet).

A total of 14 species were recorded in the tree stratum, with nine in the N-plot (2 unique, i.e., occurring only in the N-Plot) and 12 in the S-plot (5 unique). The identity of the dominants in the overstory was generally similar for the N- and S-plots, but relative proportions differed. Beech, sugar maple, and white oak made up the largest fraction of RIV in each stand (Table 1). Blackgum was also important in the S-plot, but in considerably lower abundance in the N-plot. There was a distinct trade-off between sugar maple and white oak abundance on the N- and S-plots with sugar maple being more prominent on the N-plot and white oak on the S-plot (Table 1). Overall density and basal area (Table 1) was greater on the S-plot relative to the N-plot (249 stems ha^{-1}, 34.7 m^2 ha^{-1} versus 220 stems ha^{-1}, 31.2 m^2 ha^{-1}, respectively).

Similarly, the species richness in the sapling stratum was greater on the S-plot (S=8, 2 unique) relative to the N-plot (S=10, 4 unique). Composition and abundance in this stratum differed dramatically between plots (Table 2). The N-plot was dominated by sugar maple (RIV=81%), with all other species assuming a relatively minor role (RIV<10%). In contrast, the S-plot shared dominance among several species. While beech was the primary dominant (RIV=44%), sugar maple, dogwood, and cherry were also abundant RIVs (13–17%). In contrast, in this stratum, density and basal area (Table 2) were greater on the N-plot relative to the S-plot (751 stems ha^{-1}, 1.3 m^2 ha^{-1} versus 683 stems ha^{-1}, 1.03 m^2 ha^{-1}, respectively).

The combined tree and sapling diameter distribution structure varies considerably among species and between plots (Fig. 3). The maples (sugar and red combined) exhibit a reverse-J diameter distribution. They are most abundant in the smaller size classes (2.5–25 cm) and only exist as large trees (≥50 cm) on the N-plot. Likewise, beech is most abundant in the smallest size class (2.5–10 cm) but exhibits a distinctly greater than expected (based on a negative exponential distribution) number of stems in the

Table 1
Density (DEN; stems ha^{-1}), relative density (RDEN), basal area (BA; m^2 ha^{-1}), relative basal area (RBA), and relative importance value (RIV; [RDEN+RBA]/2) of trees (stems ≥10 cm DBH) on the two study sites at Dysart Woods, Belmont County, Ohio

Site	Species	DEN	RDEN	BA	RBA	RIV
North	*Acer rubrum*	20.00	9.09	2.39	7.65	8.37
	Acer saccharum	80.00	36.36	4.63	14.83	25.60
	Fagus grandifolia	68.57	31.17	11.99	38.37	34.77
	Liriodendron tulipifera	17.14	7.79	0.64	2.04	4.91
	Nyssa sylvatica	8.57	3.90	1.45	4.64	4.27
	Prunus serotina	5.71	2.60	0.10	0.31	1.45
	Quercus alba	8.57	3.90	5.51	17.63	10.76
	Quercus rubra	5.71	2.60	4.32	13.83	8.21
	Ulmus rubra	5.71	2.60	0.22	0.70	1.65
	totals	220.00		31.24		
South	*Acer rubrum*	11.43	4.60	0.11	0.30	2.45
	Acer saccharum	57.14	22.99	1.87	5.40	14.20
	Carya ovata	2.86	1.15	0.57	1.64	1.40
	Cornus florida	14.29	5.75	0.14	0.40	3.07
	Fagus grandifolia	77.14	31.03	10.10	29.13	30.08
	Fraxinus americana	2.86	1.15	0.04	0.13	0.64
	Liriodendron tulipifera	8.57	3.45	2.61	7.53	5.49
	Nyssa sylvatica	28.57	11.49	4.25	12.26	11.88
	Quercus alba	31.43	12.64	13.63	39.31	25.98
	Quercus coccinea	2.86	1.15	0.21	0.61	0.88
	Quercus rubra	2.86	1.15	0.08	0.24	0.70
	Quercus velutina	8.57	3.45	1.05	3.04	3.24
	totals	248.57		34.68		

Table 2
Density (DEN; stems ha^{-1}), relative density (RDEN), basal area (BA; $m^2\,ha^{-1}$), relative basal area (RBA), and relative importance value (RIV; [RDEN+RBA]/2) of saplings (stems $\geq$2.5 cm and <10 cm DBH) on the two study sites at Dysart Woods, Belmont County, Ohio

Site	Species	DEN	RDEN	BA	RBA	RIV
North	*Acer rubrum*	37.14	4.94	0.10	7.48	6.21
	Acer saccharum	597.14	79.47	1.05	82.11	80.79
	Cornus florida	2.86	0.38	0.00	0.19	0.29
	*Crataegus*sp.	2.86	0.38	0.00	0.21	0.30
	Fagus grandifolia	94.29	12.55	0.09	7.37	9.96
	Ostrya virginiana	5.71	0.76	0.01	1.08	0.92
	Prunus serotina	8.57	1.14	0.02	1.34	1.24
	Ulmus rubra	2.86	0.38	0.00	0.21	0.30
	totals	751.43		1.28		
South	*Acer rubrum*	28.57	4.18	0.05	4.54	4.36
	Acer saccharum	100.00	14.64	0.20	19.11	16.88
	Amelanchier arborea	14.29	2.09	0.02	1.96	2.03
	Cornus florida	94.29	13.81	0.12	11.72	12.77
	Fagus grandifolia	308.57	45.19	0.44	42.70	43.94
	Liriodendron tulipifera	8.57	1.26	0.03	2.94	2.10
	Nyssa sylvatica	11.43	1.67	0.03	2.69	2.18
	Ostrya virginiana	5.71	0.84	0.01	0.58	0.71
	Prunus serotina	108.57	15.90	0.12	11.98	13.94
	Quercus rubra	2.86	0.42	0.02	1.77	1.09
	totals	682.86		1.03		

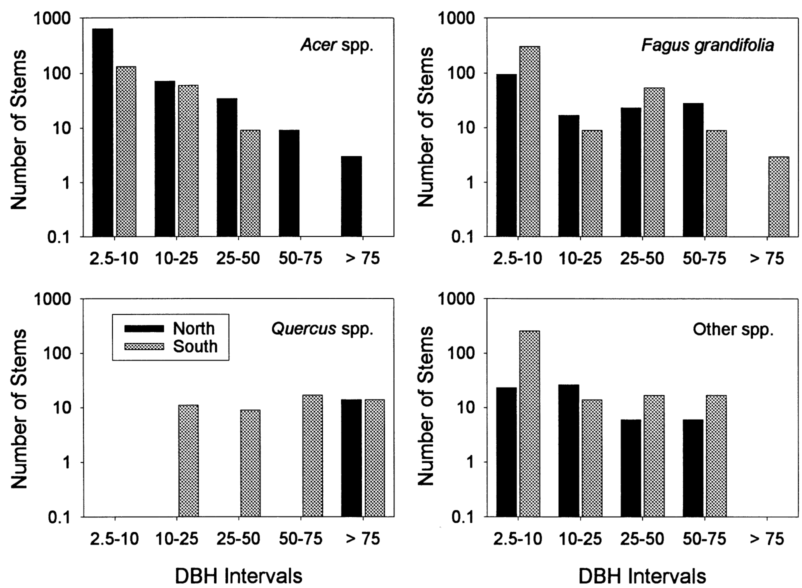

Fig. 3. Diameter distributions for live stems of *Acer* spp. (*A. rubrum* and *A. saccharum*), *Fagus grandifolia*, *Quercus* spp. (*Q. alba*, *Q. coccinea*, *Q. rubra*), and other species (primarily *Liriodendron tulipifera*, *Nyssa sylvatica*, *Fraxinus americana*, *Prunus serotina*, *Cornus florida*, *Carya ovata*). Values shown are number of stems per hectare by diameter category.

mid-size class (25–50 cm). Beech only exists in the largest size class (>75 cm) on the S-plot. Oak species present an anomaly. They are primarily found only on the S-plot and exist primarily as large individuals. Individuals in the 2.5–10 cm size class are absent in both plots. The remaining species ('others') are more or less evenly distributed among size classes (Fig. 3), but noticeably absent from the largest size class (>75 cm).

Woody plant diversity is greatest in the seedling layer (Table 3). There were 13 species (2 unique) of seedlings in the smallest size class (Class-I) in the N-plot and 19 (7 unique) in the S-plot (Table 3). Virginia creeper (*Parthenocissus quinquefolia*) and sugar maple dominated the Class-I seedling layer with the greatest number of stems in both plots. All other species, with the exception of black cherry on the S-plot, exhibited relative densities less than 5%. The overall density of Class-I seedlings was more than twice as great on the S-plot (125,000 stems ha^{-1}) as compared to the N-plot (49,000 stems ha^{-1}). The asymmetric density differential was not maintained

Table 3
Density (DEN; stems ha^{-1}) and relative density (RDEN) of Class-I (<30 cm height) and Class-II (>30 cm height, DBH<2.5 cm) seedlings on the two study sites at Dysart Woods, Belmont County, Ohio

		Class-I DEN	Class-I RDEN	Class-II DEN	Class-II RDEN
North	*Acer rubrum*	142.86	0.29		
	Acer saccharum	11428.57	23.19	3142.86	48.67
	Carya cordiformis			114.29	1.77
	Cornus florida	142.86	0.29		
	Fagus grandifolia	428.57	0.87	1142.86	17.70
	Fraxinus pensylvanica	142.86	0.29	171.43	2.65
	Lindera benzoin	142.86	0.29	628.57	9.73
	Nyssa sylvatica	571.43	1.16		
	Parthenocissus quinquefolia	30285.71	61.45	57.14	0.88
	Prunus serotina	1571.43	3.19		
	Quercus rubra	285.71	0.58		
	Ribes lacustre	857.14	1.74	342.86	5.31
	Toxicodendron radicans	2142.86	4.35	57.14	0.88
	Ulmus rubra	1142.86	2.32	800.00	12.39
	totals	49285.71		6457.14	
South	*Acer rubrum*	3428.57	2.73	57.14	0.65
	Acer saccharum	41000.00	32.65	2457.14	27.74
	Carya cordiformis			57.14	0.65
	Carya glabra	428.57	0.34		
	Celastrus scandens	1428.57	1.14	114.29	1.29
	Cornus florida	1142.86	0.91	171.43	1.94
	*Crataegus*sp.			114.29	1.29
	Fagus grandifolia	3142.86	2.50	1428.57	16.13
	Fraxinus pensylvanica	1857.14	1.48	628.57	7.10
	Lindera benzoin	2142.86	1.71	228.57	2.58
	Liriodendron tulipifera	2142.86	1.71		
	Nyssa sylvatica	714.29	0.57	285.71	3.23
	Parthenocissus quinquefolia	52857.14	42.09	57.14	0.65
	Prunus serotina	8571.43	6.83	2342.86	26.45
	Quercus alba	714.29	0.57		
	Rosa carolina	1000.00	0.80	342.86	3.87
	*Rubus*sp.	285.71	0.23		
	Toxicodendron radicans	285.71	0.23		
	Ulmus rubra	2571.43	2.05		
	Viburnum acerifolium	1142.86	0.91	342.86	3.87
	Vitis aestivalis	714.29	0.57	228.57	2.58
	totals	125571.43		8857.14	

into the next size class. Class-II seedlings were more evenly distributed in the N- (6457 stems ha^{-1}) and S-plots (8857 stems ha^{-1}). In fact, species richness also decreased with Class-II seedlings to nine species (3 uniques) in the N-plot and 15 (7 uniques) in the S-plot (Table 3). Dominance in the Class-II layer shifted to canopy species with sugar maple and beech being the most important in both plots and black cherry also being important on the S-plot only. While there are some Class-I oak seedlings on both the N- and S-plot, they are virtually unable to proceed into the next size class suggesting a strong selection force when height ≈30 cm.

The herb layer reflected diversity patterns already established in other structural layers. Regardless of measure or time period, there is greater species diversity on the S-plot relative to the N-plot (Table 4). There is also a distinct seasonal or phenological

Table 4
Mean (N=35) cover of herbaceous species in the North and South study sites sampled in mid-April, June, and August 1997 at Dysart Woods, Belmont County, Ohio

Species	North			South		
	April	June	August	April	June	August
Amphicarpa bracteata			0.13	0.14	1.97	0.40
Anemonella thalictroides					0.69	0.43
Arabis laevigata var. *laevigata*				0.17	0.49	0.11
Arisaema triphyllum		0.29	0.60			
Asarum canadense			0.29			
Aster divaricatus	0.03			5.34	9.77	13.23
Aster lateriflorus				0.84	2.69	3.29
Cardamine concatenata	13.80			1.27		
Carex digitalis					1.29	3.60
Carex laxiflora	0.01	0.43	0.11	0.14	1.71	0.74
Carex rosea				0.03	0.66	6.57
Caulophyllum thalictroides	0.40					
Cimicifuga racemosa		3.17	5.43			
Circaea lutetiana			2.29			0.83
Claytonia virginica	28.04			11.77		
Compositae (unk.)					0.51	0.57
Cryptotaenia canadensis	0.03	0.69	0.34		0.26	0.26
Desmodium glutinosum					1.11	0.60
Dryopteris carthusiana						0.17
Epifagus virginiana	0.03			0.06	0.01	
Floerkea proserpinacoides	2.86	0.60		6.76	1.06	
Galium aparine var. *aparine*	3.70	8.29		5.24	11.46	1.31
Galium concinnum					0.23	
Galium lanceolatum			0.03	0.33	1.14	1.09
Hydrophyllum macrophyllum				0.03		
Impatiens pallida	8.41	37.91	1.27		1.43	
Jeffersonia diphylla					0.06	
Lactuca canadensis var. *canadensis*					0.09	0.06
Laportea canadensis		0.34	0.17			
Lobelia inflata					0.11	
Monotropa uniflora			0.06			
Osmorhiza claytonii	11.96	17.94	10.74	3.64	11.76	9.51
Osmorhiza longistylis	1.59	5.06				
Pilea pumila			0.04		0.11	0.60
Poa cuspidata				2.84	4.63	
Poa sylvestris				0.97	0.49	
Podophyllum peltatum	5.86	10.63		0.60	1.20	
Polygonatum biflorum			0.11		0.09	0.97

Table 4 (*Continued*)

Species	North			South		
	April	June	August	April	June	August
Polygonatum pubescens					0.06	0.03
Polygonum virginianum				0.17	2.61	2.26
Potentilla simplex	0.03			0.17	0.86	0.80
Prenanthes altissima var. *altissima*				0.21	0.17	
Ranunculus arbortivus	0.09			0.20	0.20	
Ranunculus micranthus				0.09		
Sanguinaria canadensis		0.11				
Sanicula canadensis	0.09	0.11	1.00	0.09	1.97	2.80
Sedum ternatum	0.11	0.17	0.34	1.83	3.43	3.91
Smilacina racemosa					1.21	
Solidago caesia					0.51	0.54
Sphenopholis obtusa var. *major*				0.66	1.46	1.11
Trilium grandiflorum	1.59	3.31	0.43			
Uvularia sessilifolia				0.64	1.26	0.63
Viola pubescens	3.93	6.89	5.17	0.13	0.66	0.14
Species richness	19	16	18	27	38	28
Shannon diversity	1.98	1.87	1.88	2.35	2.85	2.57
Brillouin diversity	1.77	1.67	1.43	1.87	2.33	2.10
Similarity to May	100	42.44	29.89	100	43.83	30.17
Similarity to June		100	35.30		100	62.95
Similarity to August			100			100

pattern depending upon slope aspect (Table 4). The N-plot exhibited the greatest diversity in the early spring ($S=19$) and was dominated by spring ephemerals, especially spring beauty (*Claytonia virginica*). While this species was also present in abundance on the S-plot during the May sampling, dominance was shared amongst five or six species as well. The June flora was the least species rich time for the N-plot ($S=16$); most of the spring ephemerals had senesced at this point, and dominance was maintained primarily by touch-me-nots (*Impatiens pallida*) and sweet cicely (*Osmorhiza claytonii*). Touch-me-nots were virtually absent from the S-plot samplings, reflecting its drier nature. During June, the number of species on the S-plot was more than twice as great than the N-plot ($S=38$ versus 16). Sweet cicely continued to be an important dominant in both plots during the August sampling, but *Aster divaricatus* was more important on the S-plot.

Table 5
Mean (±SD) decline index (DI) for each species by plot position. The DI ranges from 0% (healthy) to 100% (dead) and is based on several crown measures (see text for details). DI values exceeding 25% are considered to be in severe decline (see text for explanation of decline categories). Data are reported for all species with N≥3 individuals with dominant or co-dominant crowns

Species	North		South	
	Mean	SD	Mean	SD
Acer rubrum	9.84	1.64	7.78	3.32
Acer saccharum	7.00	1.76	6.93	3.77
Fagus grandifolia	9.01	4.38	9.55	7.94
Liriodendron tulipifera	6.04	1.47	19.97	23.23
Nyssa sylvatica	12.33	5.27	11.13	6.34
Quercus alba	44.64	9.27	25.46	10.53
Quercus rubra	43.32	26.60		
Quercus velutina			25.14	9.45

Pairwise measures of community similarity between sample periods were relatively low (30–63%) indicating the importance of multiple sample periods for the herbaceous species (Table 4). A truly interesting discovery in this study was that 100% of the herbaceous species recorded are native to Ohio. Dozens of non-native species, some quite invasive (e.g., *Alliaria petiolata*), have existed all around the edges of the old-growth forest tracts for years but none have made it into the interior.

Results of the forest decline analysis (Table 5) clearly show all species of oak to be in severe decline on both slopes. The amount of branch dieback and crown transparency was considerable for these species (Fig. 4). Throughout the stands at Dysart Woods, the amount of white oak crown deterioration and mortality is dramatically evidenced. By comparison, the maples and other more mesic species were generally rated as healthy to trace decline (Fig. 4, Table 5).

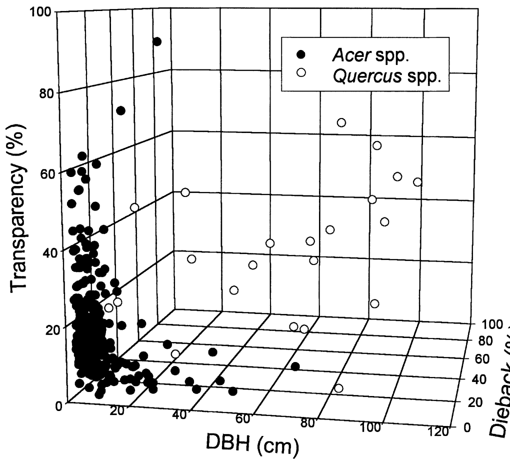

Fig. 4. Forest health assessment based on the major species (*Quercus alba*, *Q. rubra* and *Acer rubrum*, *A. saccharum*). See text for details regarding crown transparency and dieback.

The pattern of coarse woody debris (CWD) distribution and abundance (Table 6) partially reflects the structure of the live trees (Figs. 3 and 5; Table 1); both the living trees and CWD exhibit a reverse-J diameter distribution. The volume and frequency of CWD is much higher for oaks (58 and 45%) than for the maples (11 and 13%). The abundance of CWD in the later decay classes (IV & V) suggests that maple, beech, and oak have been dominants in this system for some time (Table 6, Fig. 5). In general, across all species, there appears to be consistently greater mortality and

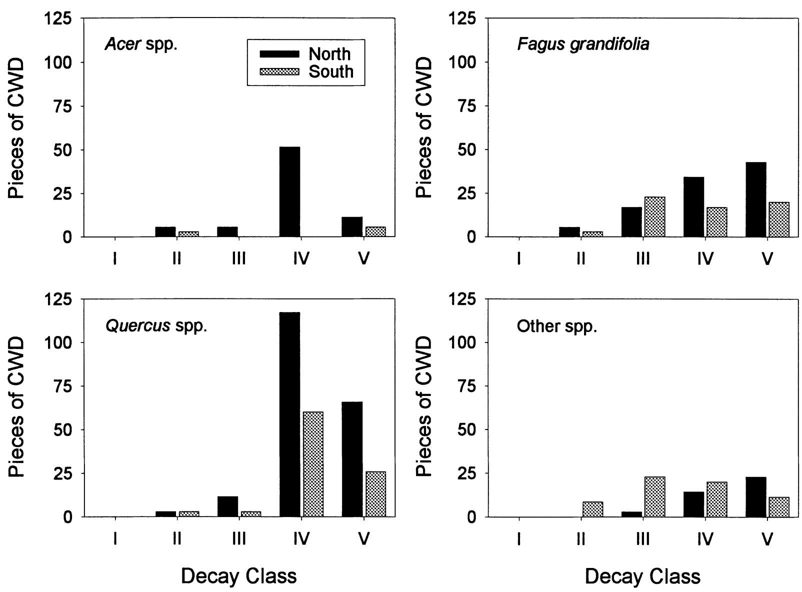

Fig. 5. Decay class distributions for logs of *Acer* spp. (*A. rubrum* and *A. saccharum*), *Fagus grandifolia*, *Quercus* spp. (*Quercus* subg. Erythrobalanus, *Q.* subg. Lepidobalanus), and other species (primarily *Liriodendron tulipifera*, *Nyssa sylvatica*, *Fraxinus americana*, *Prunus serotina*, *Carya* spp.). Values shown are numbers of pieces of coarse woody debris (mix of boles and branches) per hectare.

Table 6
Log (CWD), stump, and snag frequency (pieces ha^{-1}) and log volume (m^3 ha^{-1}) observed in the 0.35 ha N- and S-plots at Dysart Woods, a mixed-mesophytic forest in southeastern Ohio

Structural element	Species	North		South		Total	
		Frequency	Volume	Frequency	Volume	Frequency	Volume
Logs	*Acer rubrum*	11.43	3.14	2.86	0.54	14.29	3.68
	Acer saccharum	62.86	25.88	5.71	6.64	68.57	32.52
	Carya spp.			5.71	1.03	5.71	1.03
	Castanea dentata	5.71	0.72	2.86	0.57	8.57	1.29
	Cornus florida			2.86	1.39	2.86	1.39
	Fagus grandifolia	100.00	31.73	62.86	23.46	162.86	55.19
	Fraxinus spp.	5.71	6.90	11.43	5.56	17.14	12.46
	Liriodendron tulipifera	11.43	0.87			11.43	0.87
	Nyssa sylvatica	2.86	0.15	31.43	20.15	34.29	20.30
	Ostrya virginiana	5.71	3.13			5.71	3.13
	Prunus serotina	8.57	3.16	5.71	4.44	14.28	7.60
	Quercus subgen. *Erythrobalanus*	20.00	6.06			20.00	6.06
	Quercus subgen. *Lepidobalanus*	177.14	122.14	91.43	67.16	268.57	189.30
	Sassafras albidum			2.86	0.22	2.86	0.22
	total	411.42	203.88	225.72	131.16	637.14	335.04
Stumps	*Acer saccharum*			2.86		2.86	
	Carya spp.	2.86				2.86	
	Fraxinus spp.			2.86		2.86	
	Quercus subgen. *Erythrobalanus*	2.86		5.71		8.57	
	Quercus subgen. *Lepidobalanus*	8.57		2.86		11.43	
	total	14.29		14.29		28.58	
Snags	*Acer rubrum*			2.86		2.86	
	Acer saccharum	2.86				2.86	
	Cornus florida			2.86		2.86	
	Fagus grandifolia			5.71		5.71	
	Fraxinus spp.			5.71		5.71	
	Prunus serotina	2.86				2.86	
	Quercus subgen. *Erythrobalanus*			2.86		2.86	
	Quercus subgen. *Lepidobalanus*	5.71		5.71		11.43	
	total	11.43		25.71		37.14	
Logs	Decay class						
	I	0.00	0.00	0.00	0.00	0.00	0.00
	II	14.29	8.97	17.14	10.40	31.43	19.37
	III	37.14	41.60	48.57	28.74	85.71	70.34
	IV	217.14	117.66	97.14	73.71	314.28	191.37
	V	142.86	35.66	62.86	18.29	205.72	53.95

CWD production on the N-plot relative to the S-plot (Table 6, Fig. 5). This likely explains the lower density and basal area in the tree stratum of the N-plot. Notably, there is a complete absence of Class-I CWD on both plots.

By most measures, particularly the macronutrients (N, P, K, Mg), the N-plot was significantly ($P<0.05$) more fertile (Table 7). In particular, the S-plot was very low in nitrate-nitrogen and phosphorous. Aluminum concentrations were relatively high, despite a moderate pH (4–5), and resulted in low Ca:Al molar ratios for both plots (Table 7). Soil texture and organic content was similar between plots; however, qualitative observations suggest that there is a greater litter depth on the N-plot relative to the S-plot. The light environment in the understory was also dissimilar ($P<0.05$), with the S-plot receiving considerably more light.

Table 7
Mean and standard deviation (N=35) of soil parameters collected from the North and South study areas at Dysart Woods, Belmont County, Ohio

	North		South		t	P
	Mean	SD	Mean	SD		
N-NO_3 (ppm)	12.28	4.38	3.02	2.95	10.38	0.0001
P (ppm)	6.28	1.94	1.79	1.00	12.17	0.0001
K (ppm)	17.14	4.37	14.54	4.75	2.38	0.011
Ca (ppm)	81.77	136.45	42.86	108.38	1.32	0.101
Mg (ppm)	9.97	5.90	6.63	6.69	2.22	0.015
Al (ppm)	156.77	18.42	165.34	22.27	1.75	0.042
Ca:Al molar ratio	0.81	1.36	0.45	1.21	1.17	0.123
pH	4.59	0.42	4.66	0.37	0.75	0.226
% Moist	36.91	8.22	36.07	11.73	0.35	0.364
% Sand	47.11	12.28	47.32	10.63	0.07	0.469
% Clay	23.82	2.91	22.75	3.66	1.35	0.890
% Silt	29.07	12.62	29.93	10.94	0.3	0.381
% Organic matter	11.86	2.39	12.37	2.64	0.83	0.204
% Global light	3.50	1.22	5.48	3.65	3.07	0.001

5. Discussion

5.1. Composition, structure and dynamics of the overstory

As with most old-growth remnants in the eastern United States, Dysart Woods exists as an island in the landscape (McCarthy, 1997). At a local level, the matrix surrounding the woods is a mixture of agriculture land (crops and grazing), mined land, and reverting second-growth forest following clearing or logging due to Anglo settlers and later generations. At a regional level, the forest sits in the center of the Ohio River Valley, one of the most industrially polluted areas in the eastern US. Additionally, humans have modified the natural disturbance regime throughout the central hardwoods region. Clearly, humans have had an enormous impact on both the local and regional landscape.

Old-growth remains critical as a research resource. Many argue that ecologists must now turn their attention to the human-modified landscape that predominates throughout most of the world. Acknowledging this, there continues to be many important questions that can only be answered using old-growth relicts. Specifically, old-growth remains as our benchmark to understanding the natural (pre-Anglo) vegetation and disturbance regimes throughout the eastern United States. As old-growth is lost, we loose information that may be vital to our future understanding of forest ecology and management.

Sears (1925); Braun (1950), and Gordon (1969) have all made important contributions to our understanding of natural forest vegetation, both locally and regionally. White oak, sugar maple, beech, and tulip poplar have all been noted as being prominent in the natural vegetation of the region. However, the prominence of white oak in these woods, specifically, has long been a subject of question, due to its mixture with other mesic species and population age structure. Lafer and Wistendahl (1970) conducted the first quantitative analysis of Dysart Woods some 30 years prior (1967–1968) and noted the abundance of white oak, particularly in the largest size classes. They found white oak below 25 cm DBH to be virtually absent. They also noted the abundance of sugar maple and beech in the smaller diameter categories, suggesting that oak would be replaced by beech and maple. This trend continues to advance today based on the data presented here. Moreover, the failure of oak to regenerate has now become recognized on a much larger scale (e.g., Lorimer, 1993). However, this woods continues to present some anomalies which are not so simple to resolve.

The question of original white oak establishment in this woods and its current dominance remains questionable. McCarthy et al. (1987); Abrams and Downs (1990) and Lorimer (1993), among many others, have pointed to the potential change in vegetation (increased mesophytic species and decreased oaks) of the region associated with changing fire regimes over the past few centuries. Clearly, oaks appear to regenerate better on drier (upper slope), high light (SW-exposure) slopes, and prescribed fire may assist in this process. Sutherland (1997) has clearly shown that fire scars are prevalent in second growth, mixed oak forests in southern Ohio. Using dendroecological techniques, she studied the fire scars of 14 post-settlement (pith minimum was 1856) cross sections and found that fire return intervals were as frequent as 3.7 years. She also demonstrated a strong response to the fire suppression era with a dramatic suppression of fire scarring from 1930 onwards. As Sutherland acknowledges though, this provides only a record of post-settlement disturbance regimes. As far as we are aware, Buell et al. (1954) provide the only available data on presettlement fires in eastern oak forest. They found only several fire scars from the period of 1641–1711 from a single, old-growth white oak cross section obtained in a New Jersey forest.

Dendroecological analysis of Dysart Woods reveals considerable fire activity from 1731 to 1881 with only six presettlement (pre-1780) fires. These data reflect a pattern not unlike that observed by Buell et al. (1954). Fire was more prevalent in the landscape around the time of Anglo settlement. From the 1850's to present, patterns are similar (and reaffirm) those documented by Sutherland (1997). That is, a high number of fires occurred during the colonization and exploitation periods and then cessation once fire-suppression practices began. In fact, the fire suppression era may be the best explanation for the proliferation of beech in our middle diameter size classes. Overall though, the data suggest that the role of native Americans and their use of fire in upland forests may not have been as dramatic (Russell, 1983) as some have proposed (Day, 1953) and in all likelihood do not account for the abundance of white oaks now present in the woods. In fact, Kline and Cottam (1979) argue that heavily dissected topographies, similar to those present in southeastern Ohio, would be inhibitory to the regional spread of many low-intensity fires.

Examination of fire years and reconstructed PDSI values reveal no noteworthy pattern between large, negative PDSI values (i.e., dry, droughty conditions) and the presence of fire. A similar lack of correlation between climate and fire was observed by Sutherland (1997). Sutherland states that this counterintuitive disjunction between dry conditions and high fire frequency may be due to the source of fire ignition, i.e., humans not lightning. In the eastern US, electrical storms are often associated with precipitation, and, consequently, few fires are caused by lightning strikes.

Lorimer (1993) notes that 'oak regeneration failures appear to have been a widespread problem only in the last 50 years or so' with the trend 'seen most clearly. . .on mesic sites clear-cut after 1930' and many sites are now dominated by mesic species. Based on the data from Lafer and Wistendahl (1970) and the vegetation and size structure data provided in this study, white oak regeneration failure in these woods may be have originated well before Lorimer's proposed time period. Due to an absence of small and mid-size diameter cohorts, oak regeneration does not appear to have been prevalent even during the most active period of fire in these woods in the 1800s.

In addition, McCarthy and Rubino (manuscript) found that, based on growth in the region of the pith, white oak at Dysart Woods generally originated under closed canopy conditions and from multiple cohorts. This suggests that white oak might regenerate or respond to canopy gaps, however, white oak is notorious for its slow growth rate (Rogers, 1990) and simply can not compete in a gap dynamics context. Even if this were true, there is still a major anomaly in the structural data. Fig. 3 clearly shows a wave of beech ingrowth in the 25–50 cm diameter category. This could only have occurred in response to a moderate stand disturbance 75–125 years ago (perhaps the aftermath of a winter ice storm, or suppression of fire). Beech is relatively shade tolerant and capable of responding to canopy gaps (Harcombe et al., 1982). However, why was there no white oak regeneration during that same time period if it were able to respond to large canopy gaps? This suggests that gap recruitment of the original cohorts of white oak is also a poor hypothesis for why oak is abundant here.

Having ruled out most logical disturbance agents and stand dynamics issues, the most parsimonious explanation is a change in climate. Oaks are usually considered to be favored by relatively warm and dry climates, and paleocologists have long used the abundance of oak pollen as an indicator of such conditions. Based on historic pollen profiles, Ogden (1966) states that Ohio temperature and moisture have fluctuated in recent centuries. In theory, this could cause an expansion of mesophytic species on to sites that were previously warm and dry. With these climatic changes, oaks may have gained dominance in stands that were once warm and dry but are now considered too mesic for oak growth and reproduction. Ogden further states that 'drier uplands (like Dysart) were covered with an oak-hickory assemblage, and that the north slopes and protected areas supported a beech-maple forest association.' In a subsequent dendrochronological paper analyzing the climate signal of white oak at Dysart Woods, we (Rubino & McCarthy, manuscript) demonstrate that oak radial-growth rates are significantly correlated with precipitation, temperature, and drought severity. Thus, white oak, even in mesophytic forests, are climatically sensitive. Using a simple space-for-time substitution analog, this hypothesis is entirely consistent with the pattern we see in current oak regeneration and abundance on drier, SW-facing, upper slopes (reflection of niche contraction). A climate hypothesis to explain oak distribution and abundance is not new (see Kline and Cottam, 1979), however, it has been frequently ruled out as an explanation in many other regional forests (Lorimer, 1993).

5.2. Understory vegetation

Understory data is often overlooked in many forest vegetation studies. As forest managers become increasingly interested in ecosystem management, old-growth relicts will provide important benchmarks to understanding understory diversity in the managed forest ecosystem (McCarthy, 1997). Moreover, herbaceous understory data may provide insightful information regarding site quality, overstory species regeneration patterns (Gilliam and Turrill, 1993; Gilliam et al., 1995), ecosystem health, and conservation status. The effect of slope aspect on ecological pattern is known to be particularly important, especially in areas with a hilly topography such as southeastern Ohio. Numerous investigators have examined the relationship between aspect and microclimate (Wolfe et al., 1949; Cantlon, 1953), tree growth (Carmean, 1965), soil formation (Franzmeier et al., 1969), and vegetation (McCarthy et al., 1987), but understory vegetation has received considerably less attention (Olivero and Hix, 1998). Our data show a strong pattern of understory vegetation response between N- and S-facing slopes.

In addition to understory community differences associated with aspect, we also noted considerable phenological differences throughout the growing season. Most studies fail to account for the differences among early-, mid-, and late-growing season understory communities in hardwood forests (e.g., Olivero and Hix, 1998). If we are to understand the true patterns of diversity and ecological consequences in forest understories we need to develop a clearer understanding of the species distribution patterns within time as well as space.

Overall, we found a greater diversity, but lower cover, of herbs on the S-facing plot. This pattern is consistent with Grime's (Grime, 1977) model of community development under differing levels of productivity. Based on soil and light conditions, the S-plot was much less productive than the N-plot. The more intense shade generated by a more closed canopy of sugar maple on the N-facing plots may explain the lower diversity patterns there. But, while diversity was lower on the N-plot, total cover was greater (up to 90%). Herbaceous understory cover levels this great are likely to have a negative feedback on the regeneration of overstory species. In all likelihood, only the most shade tolerant of species could make it under these conditions and this may explain the high percentage of sugar maple and beech seedlings present. Interference by herbs is the most parsimonious explanation for the inability of oaks to grow in to the second size class (>30 cm height). Browsing by deer is possible, but unlikely. Ohio has a relatively high annual deer harvest and herds have not expanded to an overly detrimental size as they have in other states.

Gilliam et al. (1995) found that the abundance of woody seedlings increased with stand maturity following succession, but were unable to extrapolate this linkage past mature second-growth (70 years) due to a lack of older forests for comparison. It may be possible that the relationship between time after disturbance and the relationship between woody seedlings and herbaceous plants exhibit inverse parabolic functions. In other words, herb diversity may be greatest immediately following a disturbance, lowest at stand maturity, and increasing again at the old-growth stage of development. The abundance of woody seedlings may be the inverse of this with greatest abundance at maturity and lowest abundance at early and late stages of development. A complete replicated chronosequence stratified by aspect would be needed to test this relationship though.

5.3. Forest health

One of the most striking features of this forest is its physiognomy compared to surrounding second-growth forest. More recently, the large-scale mortality being witnessed in the overstory has become a focal point. Most of this mortality is concentrated in the oldest cohorts of white oak. Unfortunately, our understanding of the maximum age and physiological mechanisms associated with senescence in old trees is not well understood. The oldest white oaks at Dysart Woods may be reaching their physiological maximum, under forest grown conditions, at 300–350 years of age. But, these trees are unlikely dying of old age alone. Decline is believed to be a complex process.

Manion's (Manion, 1981) 'decline disease spiral' provides a good conceptual framework for understanding the proximal and ultimate reasons for forest decline and tree death at Dysart Woods. He recognizes three major sets of interacting factors that contribute to decline. 'Predisposing factors' are innate or chronic stresses that weaken a tree and permit decline to be initiated. These factors might include tree age, site quality, climate, and air pollution. 'Inciting factors' are short duration stresses or events that usually result in weakened vigor and include such things as defoliation by insects, drought, frost, and pollution. 'Contributing factors' are usually biotic agents that are directly responsible for the death of a weakened tree (e.g., canker fungi, root rot, boring insects, nematodes, saprobic decay).

There are several predisposing factors affecting the decline of white oak at Dysart Woods. First, the trees are of very great age. Unfortunately, there is little data on the absolute life time of a white oak and it remains unclear as to when white oak would die naturally in the absence of any decline factors. Secondly, given the highly dissected topography one would also expect differential decline as a function of aspect, and indeed we find lower decline values for oak on the warmer and drier SW-facing plot compared to the N-plot. Lastly, there are chronic air pollution problems in the Ohio valley (McClenahen and Dochinger, 1985; Gschwandtner et al., 1986). A dendrochronological analysis of white oak at Dysart Woods (Rubino and McCarthy, manuscript) showed a severe drought in 1988. Coincidentally, the first phenotypic signs of white oak decline appeared in the early 1990s. This drought event may have been an inciting factor that led to oak mortality in this stand based on characteristics of the proximal growth patterns (Pedersen, 1998). Most recently, there has been a proposal, by those who own the mineral rights, to mine coal using long wall mining techniques under the woods. The surface effects of such mining are variable and not readily predictable, but would likely affect hydrologic regimes and could create a chronic predisposing water shortage or inciting drought to further exacerbate oak decline. Lastly, many of the dead oaks in these stands show clear evidence of infection by the root rot *Armillaria mellea*. The two-lined chestnut borer (*Agrilus bilineatus*) is often a major contributing factor to decline in white oak following drought in this region (LeBlanc, 1998) but it was not observed in Dysart Woods.

While oak is in an obvious state of decline, all other species appear to be generally healthy and there are no forest-wide patterns of large-scale mortality or reduced growth, suggesting that the forest as a whole is in a relatively healthy condition. However, a broader concern may be the Ca:Al ratios discovered in our soils analysis. The detrimental impacts of high soil Al and low soil Ca levels on plant growth and nutrition have long been the subject of study and are well known to create considerable stress and decreased growth of forest trees (Lyon and Sharpe, 1999). For the tree species present in our study, impacts on growth and nutrient acquisition will be observed at Ca:Al ratios of 0.5–5.0 (Cronan and Grigal, 1995). All samples fell within this range. Thus, Ca:Al may be an additional predisposing factor leading to white oak mortality. Moreover, the old white oak at Dysart Woods may simply have more predisposing factors than other species and be a precursor to larger scale multi-species decline observable at later date. Increasing acid and nitrogen deposition, decreasing pH, and decreasing Ca:Al ratios appear to be a chronic (and potentially catastrophic) problem throughout the Ohio Valley

(Boerner and Sutherland, 1997; Lyon and Sharpe, 1999).

5.4. Coarse woody debris

The majority of the CWD at Dysart Woods is found in moderate to advanced stages of decay (Classes III through V). Such a distribution is common in old-growth forests (Harmon et al., 1986). Both the patchy spatial and temporal (based on decay class) woody debris distributions suggest gap-phase dynamics as the major disturbance mechanism at Dysart Woods (Harmon et al., 1986; Muller and Liu, 1991). *Quercus* subgenus *Lepidobalanus* (which would be comprised almost entirely of *Q. alba* at Dysart Woods) was the most commonly encountered woody debris type in both the N- and S-plots and represents 57% of total log volume and 42% of the total logs. Additionally, *Quercus* subgenus *Lepidobalanus* was well represented in both the stump and snag composition levels.

Such a distribution further suggests that the white oaks of Dysart may be in a state of decline. For example, examination of many of the downed logs revealed heart rot and/or *Armillaria* fungus. We hypothesize that the white oak CWD contribution will significantly increase total necromass in the stand as the remaining stems senesce. For instance, a mild ice storm that occurred following sampling resulted in a large input of snapped-off partially-rotted limbs and crowns (DLR, personal observation).

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References

Abrams, M.D., 1985. Fire history of oak gallery forests in a northeast Kansas tallgrass prairie. Am. Midl. Natl. 114, 188–191.

Abrams, M.D., Downs, J.A., 1990. Successional replacement of old-growth white oak by mixed mesophytic hardwoods in southwestern Pennsylvania. Can. J. For. Res. 20, 1864–1870.

Atwater, C., 1818. Notice of the scenery, geology, minerology, botany, etc. of Belmont County, Ohio. Am. J. Sci. 1, 226–230.

Auten, J., 1941. Notes on old-growth in Ohio, Indiana, and Illinois. Tech. Note 49., U.S.D.A. Forest Service, Columbus, OH, 8 pp.

AutoCAD, 1996. User's Guide, AutoCAD LT for Windows 95. Autodesk, Inc. San Rafael, CA, 732 pp.

Boerner, R.E.J., Cho, D.S., 1987. Structure and composition of Goll Woods, an old-growth remnant in northwestern Ohio. Bull. Torrey Bot. Club. 114, 173–179.

Boerner, R.E.J., Kooser, J., 1991. Vegetation of Drew Woods, an old-growth remnant in western Ohio, and issues of preservation. Natl. Areas. J. 11, 48–54.

Boerner, R.E.J., Sutherland, E.K., 1997. The chemical characteristics of soil in control and experimentally thinned plots in mesic oak forests along a historical deposition gradient. Appl. Soil. Ecol. 7, 59–71.

Bonham, C.D., 1989. Measurements for Terrestrial Vegetation. Wiley, New York, 338 pp.

Bormann, F.H., Likens, G.E., 1979. Patterns and Process in a Forested Ecosystem. Springer, New York, 253 pp.

Bouyoucos, G.J., 1951. A recalibration of the hydrometer method for making mechanical analysis of soil. Agron. J. 43, 434–437.

Braun, E.L., 1950. Deciduous Forests of Eastern North America. Hafner Press, New York, 596 pp.

Bryant, W.S., 1987. Structure and composition of the old-growth forests of Hamilton County, Ohio and environs. In: Hay, R.L., Woods, F.W., DeSelm, H.R. (Eds.), Proceedings of the 6th Central Hardwood Forest Conference. USDA Forest Service, Knoxville, TN, pp. 317–324.

Buell, M.F., Buell, H.F., Small, J.A., 1954. Fire in the history of Mettler's Woods. Bull. Torrey Bot. Club 81, 253–255.

Caldwell, J.A., 1880. History of Belmont and Jefferson Counties, Ohio. Historical Publishing Company, Wheeling, WV, 611 pp.

Canham, C.D., 1995. GLI/C Color fish-eye photo analysis, v 3.1. Institute of Ecosystem Studies, Millbrook, NY.

Cantlon, J.E., 1953. Vegetation and microclimates on north and south slopes of Cushetunk Mountain, New Jersey. Ecol. Monogr. 23, 241–270.

Carmean, W.H., 1965. Black oak site quality in relation to soil and topography in southeastern Ohio. Soil. Sci. Soc. Proc. 29, 308–312.

Carter, M.R., 1993. Soil Sampling and Methods of Analysis. Lewis Publishers, Boca Raton, FL, 823 pp.

Cook, E.R., Meko, D., Stahle, D.W., Cleaveland, M., 1999. Reconstruction of past drought across the coterminous united states from a network of climatically sensitive tree-ring data. National Oceanic and Atmospheric Administration/National Geophysical and Climate Data Centers. http://www.ngdc.noaa.gov/paleo/usclient2.html.

Cox, G.W., 1990. Laboratory Manual of General Ecology, 6th Edition. Brown Publishers, Dubuque, IA, 251 pp.

Cronan, C.S., Grigal, D.F., 1995. Use of calcium/aluminum ratios as indicators of stress in forest ecosystems. J. Environ. Qual. 24, 209–226.

Davison, S.E., Forman, R.T., 1982. Herb and shrub dynamics in a mature oak forest, a 30 year study. Bull. Torrey Bot. Club. 109, 64–73.

Day, G.M., 1953. The Indian as an ecological factor in the northeastern forest. Ecology 34, 329–426.

Denevan, W.M., 1992. The pristine myth, the landscape of the Americas in 1492. Ann. Assoc. Am. Geogr. 34, 329–346.

Fenneman, N., 1938. Physiography of Eastern United States. McGraw Hill, New York, 714 pp.

Franzmeier, D.P., Pederson, E.J., Longwell, T.J., Byrne, J.G., Losche, C.K., 1969. Properties of some soils in the Cumberland plateaus as related to slope aspect and topography. Soil Sci. Soc. Am. Proc. 33, 755–761.

Gilliam, F.S., Turrill, N.L., 1993. Herbaceous layer cover and biomass in a young versus mature stand of central Appalachian hardwood forest. Bull. Torrey Bot. Club 120, 440–450.

Gilliam, F.S., Turrill, N.L., Adams, M.B., 1995. Herbaceous-layer and overstory species in clear-cut and mature central Appalachian hardwood forests. Ecol. Appl. 5, 947–955.

Gleason, H.A., Cronquist, A., 1991. Manual of Vascular Plants of Northeastern United States and Adjacent Canada. The New York Botanical Garden, Bronx, NY, 910 pp.

Gschwandtner, G., Gschwandtner, K., Eldridge, K., Mann, C., Mobley, D., 1986. Historic emissions of sulfur and nitrogen oxides in the United States from 1900 to 1980. J. Air Pollut. Control Assoc. 36, 39–149.

Goebel, P.C., Hix, D.M., 1996. Development of mixed-oak forests in southeastern Ohio, a comparison of second-growth and old-growth forests. For. Ecol. Manage. 84, 1–21.

Gordon, R.B., 1969. The natural vegetation of Ohio in pioneer days. Bull. of the Ohio Biological Survey, New Series, Vol. 3, No. 2. The Ohio State University, Columbus, OH, 113 pp.

Grime, J.P., 1977. Evidence for the existence of three primary strategies in plants and its relevance to ecological and evolutionary theory. Am. Natl. 11, 1169–1194.

Hach Co., 1990. N-trak soil test kit manual. Loveland, CO, 16 pp.

Harcombe, P.A., White, B.D., Glitzenstein, J.S., 1982. Factors influencing the distribution and and first-year survivorship of a cohort of beech (*Fagus grandifolia* Ehrl). Castanea 47, 148–157.

Harmon, M.E., Franklin, J.F., Swanson, F.J., Sollins, P., Gregory, S.V., Lattin, J.D., Anderson, N.H., Cline, S.P., Aumen, N.G., Sedell, J.R., Lienkaemper, G.W., Cromack Jr., K., Cummins, K.W., 1986. Ecology of coarse woody debris in temperate ecosystems. Adv. Ecol. Res. 15, 133–302.

Huston, M.A., 1994. Biological Diversity, The Coexistence of Species on Changing Landscapes. Cambridge University Press, Cambridge, 681 pp.

Hutchins, R.L., Hill, J.D., White, E.H., 1976. The influence of soils and microclimate on vegetation of forested slopes in eastern Kentucky. Soil Sci. 121, 234–241.

Juday, G.P., 1988. Old-growth forests and natural areas, an introduction. Natl. Areas J. 8, 3–6.

Kline, V.M., Cottam, G., 1979. Vegetation response to climate and fire in the Driftless Area of Wisconsin. Ecology 60, 861–868.

Kovach, W.L., 1999. MVSP — A Multivariate Statistical Package for Windows, Version 3.10b. Kovach Computing Services, Pentraeth, Wales, UK, 127 pp.

Knepper, G., 1989. Ohio and its People. The Kent State University Press, Kent, OH, 508 pp.

Lafer, N.G., Wistendahl, W.A., 1970. Tree composition of Dysart Woods, Belmont County, Ohio. Castanea 35, 302–308.

LeBlanc, D.C., 1998. Interactive effects of acidic deposition, drought, and insect attack on oak populations in the midwestern United States. Can. J. For. Res. 28, 1184–1197.

Lorimer, C.G., 1993. Causes of the oak regeneration problem. In: Loftis, D., McGee, C.E. (Eds.), Oak Regeneration: Serious Problems, Practical Recommendations. Gen. Tech. Report SE-84, U.S.D.A. Forest Service, Asheville, NC, pp. 14–39.

Lyon, J., Sharpe, W.E., 1999. An assessment of the Ca:Al ratios of selected Pennsylvania forest soils. Water, Air, Soil Pollut. 109, 53–65.

Magurran, A.E., 1988. Ecological Diversity and its Measurement. Princeton University Press, Princeton, NJ, 179 pp.

Manion, P.D., 1981. Tree Disease Concepts. Prentice Hall, Englewood Cliffs, NJ, 399 pp.

Manion, P.D., Lachance, D., 1992. Forest decline concepts, an overview. In: Manion, P.D., Lachance, D. (Eds.), Forest Decline Concepts. The American Phytopathological Society, St. Paul, MN, pp. 181–190.

Martin, W.H., 1975. The Lilley Cornett Woods, a stable mixed mesophytic forest in Kentucky. Bot. Gaz. 136, 171–183.

Martin, W.H., 1992. Characteristics of old-growth mixed mesophytic forests. Natl. Areas J. 12, 127–135.

McCarthy, B.C., 1995. Eastern old-growth forests. Ohio Woodland J. 1, 8–10.

McCarthy, B.C., 1997. Lab Protocols for the Testing of Eastern Deciduous Forest Soils. URL, http://www.plantbio.ohiou.edu/epb/soils/soils.htm.

McCarthy, B.C., Bailey, D.R., 1996. Composition, structure, and disturbance history of Crabtree Woods, an old-growth forest of western Maryland. Bull. Torrey Bot. Club 123, 350–365.

McCarthy, B.C., Bailey, R.R., 1994. Distribution and abundance of coarse woody debris in a managed forest landscape of the central Appalachians. Can. J. For. Res. 24, 1317–1329.

McCarthy, B.C., Vierheller, T.L., Wistendahl, W.A., 1984. Species ordination of upper slope oak-hickory stands of southeastern Ohio. Bull. Torrey Bot. Club 111, 56–60.

McCarthy, B.C., Hammer, C.A., Kauffman, G.L., Cantino, P.D., 1987. Vegetation patterns and structure of an old-growth forest in southeastern Ohio. Bull. Torrey Bot. Club 114, 33–45.

McClenahen, J.R., Dochinger, L.S., 1985. Tree ring response of white oak to climate and air pollution near the Ohio River valley. J. Environ. Qual. 14, 274–280.

McLaughlin, D., Gizyn, W., McIvern, W., Kinch, C., 1992. A quantitative tree crown rating system for deciduous forest health surveys, some results from Ontario. In: Manion, P.D., Lachance, D. (Eds.), Forest Decline Concepts. The American Phytopathological Society, St. Paul, MN, pp. 137–154.

Millers, I., Lachance, D., Burkman, W.G., Allen, D.C., 1991. North American sugar maple decline project, organization and field methods. Gen. Tech. Rep. NE-154, U.S.DA. Forest Service, Radnor, PA, 26 pp.

Mueller-Dombois, D., Ellenberg, H., 1974. Aims and Methods of Vegetation Ecology. Wiley, New York, 547 pp.

Muller, R.N., 1982. Vegetation patterns in the mixed mesophytic forest of eastern Kentucky. Ecology 63, 1901–1917.

Muller, R.N., Liu, Y., 1991. Coarse Woody debris in an old-growth forest on the Cumberland plateau, southeastern Kentucky. Can. J. For. Res. 21, 1567–1572.

National Climatic Data Center/National Oceanic and Atmospheric Administration, 1999. CLIMVIS Information, Selection criteria for displaying period of record. http://www.ncdc.noaa.gov/onlineprod/drought/xmgrg3.html.

Odum, E.P., 1969. The strategy of ecosystem development. Science 164, 262–270.

Ogden III, J.G., 1966. Forest history of Ohio. I. Radiocarbon dates and pollen stratigraphy of Silver Lake, Logan County, Ohio. Ohio J. Sci. 66, 387–400.

Oliver, C.D., Larson, B.C., 1990. Forest Stand Dynamics. McGraw Hill, New York, 467 pp.

Olivero, A.M., Hix, D.M., 1998. Influence of aspect and stand age on ground flora of southeastern Ohio forest ecosystems. Plant Ecol. 139, 177–187.

Palmer, W. C., 1965. Meteorological drought. Res. Pap. No. 45. U.S. Dept. of Commerce Weather Bureau. Washington, D.C., 58 pp.

Panshin, A.J., deZeeuw, C., 1980. Textbook of Wood Technology, 4th Edition. McGraw-Hill, New York, 722 pp.

Parker, G.R., 1989. Old-growth forests of the Central Hardwood Region. Natl. Areas. J. 9, 5–11.

Parker, G.R., Leopold, D.J., Eichenberger, J.K., 1985. Tree dynamics in an old-growth, deciduous forest. For. Ecol. Manage. 11, 31–57.

Patterson III, W.A., Sassaman, K.E., 1988. Indian fires in the prehistory of New England. In: Nicholas, G.P. (Ed.), Holocene Human Ecology in Northeastern North America. Plenum Press, New York, pp. 107–135.

Peacefull, L., 1996. The land. In: Peacefull, L. (Ed.), A Geography of Ohio. The Kent State University Press, Kent, OH, pp. 3–15.

Pedersen, B.S., 1998. The role of stress in the mortality of midwestern oaks as indicated by growth prior to death. Ecology 79, 79–93.

Robison, S.A., McCarthy, B.C., 1999. Potential factors affecting the estimation of light availability using hemispherical photography in oak forest understories. J. Torrey Bot. Soc. 126, 344–349.

Rogers, R., 1990. *Quercus alba*, White Oak. In: Burns, R.M., Honkala, B.H. (Eds.), Silvics of North America, Vol. 2. Hardwoods. Agric. Handbook 654, U.S.D.A. Forest Service, Washington, DC, pp. 605–613.

Rubel, N., Jenny P.C., Plunkett, M.K., 1981. Soil Survey of Belmont County, Ohio. U.S.D.A., Soil Conservation Service. Washington, DC, 178 pp.

Russell, E.W.B., 1983. Indian-set fires in the forests of the northeastern United States. Ecology 64, 78–88.

Schmidlin, T.W., 1996. Climate and weather. In: Peacefull, L. (Ed.), A Geography of Ohio. The Kent State University Press, Kent, OH, pp. 16–30.

Sampson, H.C., 1930. The mixed mesophytic forest community of northeastern Ohio. Ohio J. Sci. 30, 358–367.

Sears, P.B., 1925. The natural vegetation of Ohio. I. Map of the virgin forest. Ohio J. Sci. 25, 139–149.

Smith, K.T., Sutherland, E.K., 1999. Fire-scar formation and compartmentalization in oak. Can. J. For. Res. 29, 166–171.

Stokes, M.A., Smiley, T.L., 1968. An Introduction to Tree-Ring Dating. University of Chicago Press, Chicago, 73 pp.

Sutherland, E.K., 1997. History of fire in a southern Ohio second-growth mixed-oak forest. In: Pallardy, S.G., Cecich, R.A., Garrett, H.E., Johnson, P.S. (Eds.), Proceedings of the 11th Central Hardwood Forest Conference, 23–26 March, Columbia, MO. Gen. Tech. Report NC-188, U.S.D.A. Forest Service, St. Paul, MN, pp. 172–183.

Trautman, M.B., 1977. The Ohio Country from 1750 to 1977 — A Naturalist's View. Biological Notes No. 10. Ohio Biological Survey. The Ohio State University, Columbus, OH, 25 pp.

United States Geological Survey, 1972. Armstrong Mills Quadrangle, Ohio 7.5 Minute Topographic Series (Topographic) Map. United States Department of the Interior, Geological Survey, Reston, VA.

Walker, J.E., 1983. The travel notes of Joseph Gibbons, 1804. Ohio Hist. 92, 96–146.

Walters, G.M., McCarthy, B.C., 1997. Forest decline and tree mortality in a southeastern Ohio Oak-Hickory forest. Ohio J. Sci. 97, 5–9.

Whitney, G.G., 1987. Some reflections on the value of old-growth forests, scientific and otherwise. Natl. Areas J. 7, 92 –99.

Wistendahl W.A., 1971. The Dysart Woods. Ohio Woodlands: Conservation in Action 10, 12–13.

Wilcove, D.S., 1987. From fragmentation to extinction. Natl. Areas J. 7, 23–29.

Wolfe, J.N., Wareham, R.T., Schofield, H.T., 1949. Microclimates and Macroclimate of Neatoma, A Small Valley in Central Ohio. Ohio Biol. Survey, Bull. 41, The Ohio State University, Columbus, OH, 267 pp.